

NovaBank Banking Transaction Analysis – Executive Summary

Prepared for: Chief Customer Officer / NovaBank

1. Business Context & Problem

NovaBank is a mid-sized retail bank, operating across Maharashtra and Karnataka, serving approximately 85,000 retail customers through branch and digital channels. The bank has been actively expanding its digital presence across UPI, mobile banking, and online transfers.

The Chief Customer Officer, Mr. Rajesh Mehta, observed that despite growing transaction volumes, customer engagement and retention were not improving. He needed a data-driven understanding of how customer transaction behavior was evolving, and why so many customers appeared to be disengaged.

2. Analytical Approach

For this analysis, a representative sample of 297,129 transactions, 6,356 accounts, and 5,000 customers across 10 branches between January 2022 and December 2023 was analyzed to understand customer behavior, engagement, and retention patterns.

- **SQL** for data cleaning and validation across all four tables
- **Python** for feature engineering — RFM scoring, lifecycle staging, cohort preparation, behavioral metrics
- **Tableau** for 5 executive dashboards covering customer, transaction, financial, and cohort analysis

3. Key Performance Indicators

40.59% Retention Rate <i>Industry benchmark: 75–85%</i>	54.42% Never Transacted <i>2,721 of 5,000 customers</i>	82.68% Transaction Success Rate <i>Industry benchmark: 97–98%</i>	₹ -2Cr Net Cash Flow <i>Outflow exceeds inflow</i>
---	---	---	--

Primary KPI: Customer Retention Rate — the percentage of established customers actively transacting with NovaBank. Retention was treated as the primary success metric, with activation, transaction reliability, and engagement analyzed as supporting drivers.

4. Key Insights

1. NovaBank has an activation problem, not a retention problem.

Customers who complete their first transaction in Month 1 retain at 65–91% for up to 12 months. The problem is that only around 40% of customers ever activate.

2. 54.42% of customers never transacted after joining.

2,721 out of 5,000 customers have zero transaction history. The bank is spending acquisition budget on customers who generate zero revenue.

3. Digital adoption is strong but engagement is not.

85.1% of transactions are digital and MATU grew 84x in 2 years, yet the retention rate remains at 40.59%. Growth is not translating into loyalty.

4. 1 in 6 transactions fails.

NovaBank's transaction failure rate of 16.5% is significantly above the industry benchmark (2–3%). Failed transactions damage customer trust and create friction during the most important stage of the customer journey i.e early engagement.

5. Customers are financially capable but disengaged.

99.4% of portfolio balance sits in active accounts. This is not a financial capacity problem — it is an engagement and activation problem.

6. Only 18 customers are High Value.

Only 18 customers qualify as High Value based on RFM segmentation, while 94.3% of customers fall into the Low Engagement segment. NovaBank's portfolio health depends disproportionately on a very small group of customers.

7. 2023 cohorts show declining activation quality.

Month 1 retention has dropped compared to 2022 cohorts, suggesting onboarding quality is deteriorating as acquisition scales.

5. Key Recommendations

- **R1 — Improve Customer Activation (High):** Launch first-transaction cashback and UPI rewards within 30 days of account opening.
- **R2 — Reduce Transaction Failures (High):** Audit payment gateways and fix UPI and Debit Card failure points to bring failure rate under 5%.
- **R3 — Increase Product Engagement (Medium):** Introduce cashback campaigns, auto-pay for bills, and monthly engagement nudges for low-activity customers.
- **R4 — Cross-Sell to Salaried Customers (Medium):** Offer pre-approved loan products to customers with consistent salary credits.
- **R5 — Protect High Value Customers (High):** Assign dedicated relationship managers to the 18 High Value customers with premium servicing.

Immediate Priority (0-3 Months): Improve activation & Reduce transaction failures.

Mid-Term Priority (3-6 Months): Product engagement, Cross-sell opportunities, High-value customer retention.

6. Business Impact

This analysis gives Mr. Mehta a data-driven foundation to act on NovaBank's engagement crisis. The findings directly enable:

- **Targeted activation campaigns** focused on the first 30 days — where the data shows the highest leverage.
- **Operational improvements** to the payment system that will recover thousands of failed transactions annually.
- **Revenue protection** through dedicated management of the 18 High Value customers who disproportionately support portfolio health.
- **Smarter acquisition** by understanding that the problem is not getting customers in — it is getting them started.

7. Conclusion

NovaBank's challenge is not customer acquisition but customer activation. The analysis shows that customers who become active early retain strongly, while poor onboarding, transaction friction, and weak engagement are limiting long-term value. By improving first-month activation and reducing transaction failures, NovaBank can materially improve retention and customer lifetime value.